Format Unified

You are given a question and a solution, and your task is to translate the reasoning process from mathematical language into a style of natural language reasoning.
Instructions:
- Convert the formal mathematical solution into a **natural language explanation**.
- Use "Step X:" to separate each step.
- End the solution with "†Answer: [final answer]".
- Ensure the explanation is clear, concise, and free of formal mathematical symbols unless absolutely necessary.
- **You must strictly **follow the provided solution steps and final answer**. **Do not modify, correct, or reinterpret** any part of the solution.**

Example:
Question:
Given that OB is perpendicular to OA, line CD passes through point O, and ∠AOC = 20°, then ∠BOD = ?
Formal Solution:
Step 1 (Definition of Perpendicular): ∵ OB ⊥ OA, ∴ ∠BOA = 90°.
Step 2 (Given condition): ∵ ∠AOC=20°, ∴ ∠BOC = 90° - ∠AOC = 90° - 20° = 70°.
Step 3 (Supplementary angles property): ∠BOD = 180° - ∠BOC = 180° - 70° = 110°.
Answer: \\boxed{110}

Your Response:
Step 1: Since OB is perpendicular to OA, we have that angle BOA is 90 degrees.
Step 2: Given that angle AOC is 20 degrees, it follows that angle BOC is 90 degrees minus angle AOC, which is 70 degrees.
Step 3: Using the property of supplementary angles, angle BOD is 180 degrees minus angle BOC, which is 110 degrees.
†Answer: 110

Now, let's begin!
Question:
{QuestionText}
Solution:
{SolutionText}
** Your Response:

Figure 14: Format unify prompt for mathematical and symbolic reasoning style data.

divided into two categories: (i) the lack of visual condition alignment. Solutions in this category exhibit inconsistencies in reasoning regarding basic visual factors such as edge relationships, coordinate values, and theorem applications, revealing deficiencies in the pretraining phase, as shown in Figure 20. (ii) the exploitation of shortcut patterns. These rollouts do not go through key steps but are guided directly to the correct answer after basic descriptions due to the high correlation between image features and problem-solving patterns during pretraining or SFT, as shown in Figure 21. Therefore, PS-GRPO suppresses the advantageous direction brought by these erroneous actions through the sensitivity of the PRM in online RL for error identification. This leads to a more optimal paradigm that combines outcome rewards with process reward-based penalties.

H.4 Cases on How Naive Process Reward Modeling Fails

In this section, we elaborate on the two fundamental flaws of process reward guided RL mentioned in Section 4 and present some cases for illustration. In online RL, models can easily recognize the patterns for obtaining process rewards, leading to conservative analyses and concise responses as they sidestep PRM scrutiny. As shown in Figure 22, we have observed that the model often follows a distinct reasoning pattern. They initially read and analyze the given conditions comprehensively, but then make incorrect decisions based on this analysis, leading to wrong answers. This indicates that

Double Checking

You are tasked with evaluating whether a provided solution **aligns precisely** with a given standard answer. Your evaluation must strictly adhere to the following criteria:

Solution Fidelity

- **Certainty**: The solution must be free of any elements of doubt or speculative reasoning. Avoid using phrases that imply uncertainty or hypothetical reasoning, such as “more conditions are needed”, “the answer seems questionable”, “let us assume”, “the provided solution seems wrong” or any similar expressions. The solution should be presented with confidence and clarity, reflecting a definitive and well-supported conclusion.

Solution Consistency

- **Alignment with Standard Answer**: The final conclusion of the solution must match the given standard answer exactly. The reasoning process should directly and unequivocally lead to the provided answer without deviation. Ensure that there are no discrepancies between the solution's conclusion and the standard answer.

Evaluation Process

1. **Analyze the Solution**: Carefully review the logical steps and reasoning used in the solution to ensure they are sound and lead directly to the conclusion.
2. **Compare with Standard Answer**: Verify that the conclusion of the solution is **identical to the standard answer** provided.
3. **Provide Judgment**: Based on your analysis, offer a clear and concise judgment

finally:

- Respond with “**Judgment: yes**” if the solution‘s conclusion is consistent with the given standard answer.
- Respond with “**Judgment: no**” if the solution‘s conclusion does not match the standard answer.

Example Structure

- **Question**:

{QuestionText}

- **Standard Answer**:

{AnswerText}

- **Solution Answer**:

{SolutionText}

Your Response

Figure 15: Double-checking prompt for ensuring high-quality and appropriate trajectories in synthesized CoT reasoning data.

when explicitly modeling process rewards, models can easily focus on processes that seem "correct" in isolation. However, these processes may not be genuinely helpful for the final outcome and instead may lead the model to prioritize high process rewards over accuracy.

Misinterpretation Insertion for Geometry Problem

You are given a geometry problem with an image and its solution. Your task is to introduce errors into the solution by misreading the diagram and generate an incorrect answer.
Instructions:

- Your response involves three Stages:
  ** Stage 1: Analyze the Correct Solution **: Identify where in the solution diagram information is extracted. Note these areas as **potential points for introducing misinterpretations**.
  ** Stage 2: Introduce a Misinterpretation **: Choose one of the identified points from Stage 1 and **alter it to create a misleading scenario**. Integrate this misinterpretation naturally into the solution.
  ** Stage 3: Continue Reasoning **: Based on the misinterpretation from Stage 2, **continue the reasoning** process to derive an incorrect final answer.

- Ensure the misreading is **naturally integrated** without explicit statements about making a misinterpretation. When describing corrupted solution, avoid showing the misleading thought process. You must respond in a typical problem-solving style.
- Remember that the <pos> and <neg> tags only need to appear at the end of each Step X. Do not repeat them between Step X and Step X+1, even if the reasoning process for that step spans many lines.
- Remember that once a step is marked as <neg>, all subsequent steps are also considered <neg>.
- '†Answer:' is the final answer, and it should not be tagged.
- You **cannot** tag an unaltered step with <neg> without misinterpreting it. You must misinterpret the step if you are tagging it with <neg>.
- Words that imply misinterpretation, such as **'misread' 'incorrect' or 'incorrectly assume'** must not appear in your response under any circumstances.
- The misinterpretation action in Stage 3 should be **consistent** with what was planned in Stage 2.

Question:
{QuestionText}

Solution:
{SolutionText}

** Your response:

Figure 16: Misinterpretation insertion for geometry-related problems.